

**EC23S01 NUMERICAL AND SIGNAL
PROCESSING PRACTICE THROUGH PYTHON
PROJECT REPORT
WIFI RSSI MONITOR**

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SUBJECT/CODE: EC23S01-NUMERICAL AND SIGNAL
PROCESSING THROUGH PYTHON

**Certified to the bonafide record of Project work done
S.SAMUEL , S.J.PRASANTH AND M.AMIRTHAVARSINI in the
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ABSTRACT

This project focuses on the design and implementation of a Wi-Fi RSSI (Received Signal Strength Indicator) Monitoring System using the ESP32 microcontroller operating in promiscuous mode. The ESP32, equipped with an integrated Wi-Fi transceiver, is configured to capture all wireless packets transmitted within its range. Each captured packet contains essential metadata such as the MAC (Media Access Control) address and the RSSI value, which are used to identify and monitor nearby Wi-Fi-enabled devices. The project demonstrates the practical application of RSSI measurements for proximity-based systems. Since RSSI reflects the received signal power, it provides a relative indication of the distance between the ESP32 and the transmitting device. Although not suitable for precise distance estimation, the data can effectively indicate whether a device is approaching or moving away.

WIFI RSSI MONITOR

AIM:

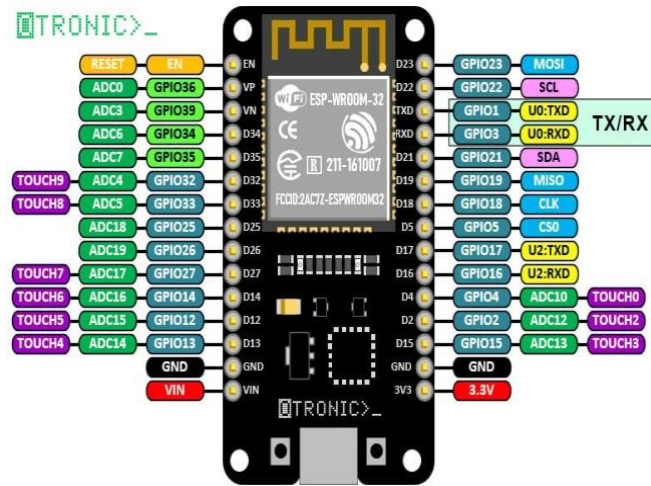
The main aim of this project is to develop an ESP32-based Wi-Fi RSSI Monitoring System capable of detecting nearby Wi-Fi devices through their MAC addresses and continuously measuring their Received Signal Strength Indicator (RSSI) values.

OBJECTIVES:

- To configure the ESP32 microcontroller in promiscuous mode for Wi-Fi packet sniffing and data capture.
- To identify and list nearby Wi-Fi devices by extracting their unique MAC addresses from captured packets.
- To measure and display the RSSI values of detected devices to determine their relative proximity.
- To enable real-time visualization of RSSI variation using the Arduino Serial Plotter.
- To explore IoT and wireless monitoring applications using the collected RSSI data.

COMPONENT DESCRIPTION:

1. ESP32 Development Board



The ESP32 is a powerful and versatile microcontroller developed by Espressif Systems. It comes with integrated Wi-Fi and Bluetooth capabilities, making it ideal for IoT and wireless communication projects. The board features a dual-core 32-bit processor, running at up to 240 MHz, with built-in SRAM and flash memory for program storage. It has multiple GPIO pins, ADC/DAC channels, PWM outputs, UART, SPI, and I²C interfaces, allowing it to connect with various sensors and devices. The ESP32 can operate in promiscuous mode, which enables it to capture Wi-Fi packets from surrounding networks for analysis. This feature is used in this project to extract MAC addresses and RSSI values.

Key Features:

- Dual-core 32-bit LX6 processor (up to 240 MHz)
- Built-in Wi-Fi (802.11 b/g/n) and Bluetooth 4.2
- 30–38 GPIO pins depending on the board model
- Supports UART, SPI, I²C, PWM, ADC, DAC interfaces

2. B-TYPE Data Cable



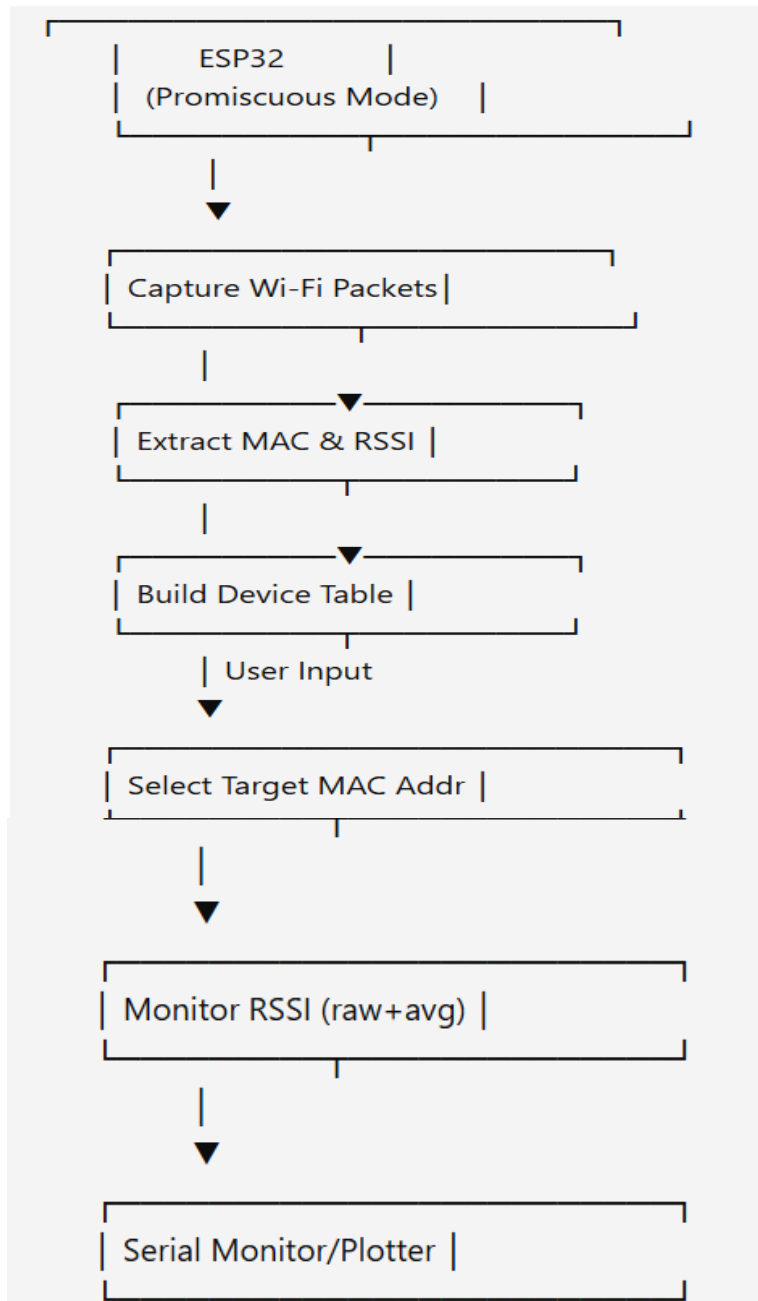
A USB Type-B Data Cable is used to connect the ESP32 development board to a computer for both power supply and data transfer. It enables the uploading of programs from the Arduino IDE to the ESP32 and also provides a serial communication link for monitoring data output.

The cable has a Type-A connector (for the computer) and a Type-B or Micro-B connector (for the ESP32 board, depending on model). It ensures stable communication and continuous 5V power during operation.

Key Features:

- Used for programming and serial communication with ESP32
- Provides 5V DC power from the computer's USB port
- Plug-and-play connection with Arduino IDE
- Supports reliable and fast data transmission

BLOCK DIAGRAM:



WORKING PRINCIPLE:

The working of the ESP32-based Wi-Fi RSSI Monitor is divided into three major phases: scanning, device selection, and monitoring.

In the scanning phase, the ESP32 microcontroller operates in promiscuous mode, allowing it to capture all nearby Wi-Fi packets transmitted within its frequency range (channels 1–13). Unlike normal Wi-Fi operation, where a device only processes packets intended for it, promiscuous mode enables the ESP32 to listen to every packet in the air. Each captured packet contains important information such as the source and destination MAC addresses, RSSI (Received Signal Strength Indicator), and channel number. The program extracts the MAC address and RSSI values from these packets and builds a device table that lists all the unique devices detected along with their average signal strengths. This step helps identify which Wi-Fi-enabled devices are present in the surroundings.

Next comes the device selection phase, where the user interacts with the Arduino Serial Monitor to view the list of all detected MAC addresses. Each device is displayed with an index number for easy identification. The user can then select a specific target device by entering the corresponding index number. Once a device is chosen, the ESP32 filters incoming packets and processes only those belonging to the selected MAC address, ignoring all others.

In the monitoring phase, the ESP32 continuously reads the RSSI values of the chosen device's packets in real time. The measured signal strengths are displayed on the Serial Monitor as numerical data, and at the same time, a dynamic graph can be viewed on the Arduino Serial Plotter, showing how the signal strength varies over time.

APPLICATIONS:

1. Proximity Detection

- Detects when a known device (e.g., smartphone, laptop) enters or leaves a specific area.
- Can be used to trigger actions like unlocking doors or turning on lights.

2. IoT Smart Systems

- Automate home or industrial devices based on the presence of a Wi-Fi-enabled device.
- Example: Switching appliances on/off when a user is nearby.

3. Signal Strength Analysis

- Study how Wi-Fi signals vary with distance, obstacles, and interference.
- Helps in optimizing router placement or wireless network performance.

4. Network Monitoring

- Observe and log the presence of Wi-Fi devices in an area.
- Useful for security audits or identifying unauthorized devices.

5. Educational Purposes

- Demonstrates wireless communication concepts and packet sniffing techniques.
- Provides hands-on learning for IoT, networking, and embedded system students.

RESULT:

The project achieved all its objectives by enabling Wi-Fi packet sniffing, MAC address identification, and real-time RSSI monitoring. The ESP32 proved to be an effective platform for developing a compact, educational, and practical Wi-Fi monitoring tool.